

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Scaler
Manufacturer: Extron
Firmware Version: 1.02
Model(s): IN1804, IN1804 DI, IN1804 DO, IN1804 DI/DO

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.04.0001-b002	2.3.1

Version History

Module Version	Date	Notes
1_2_4_0	4/7/2020	Added IN1804 DI/DO model. Tested on model IN1804 DI/DO. Changed Ethernet protocol to SSH.
1_2_1_0	6/21/2019	- Added 576p (50Hz) value to OutputResolution command. - Added ContactStatus and Tally commands - Added additional presets - Renamed all Output qualifier values to '1A' and '1B' - Fixed AudioMute, HDCPInputStatus, HDCPOutputStatus, TestPattern, and VideoMute status. - Renamed commands: - UserPresetRecall → InputPresetRecall - UserPresetSave → InputPresetSave - Renamed states: - AudioFormat - LPCM-2Ch Digital → LPCM-2Ch - Multi-Ch Digital → Multi-Ch - LPCM-2Ch Digital Auto Analog → LPCM-2Ch Auto

Global Scriptor Module Communication Sheet

		▪ Multi-Ch Digital Auto Analog → Multi-Ch Auto ○ AutoSwitchMode ▪ Input Memory Priority → Last Connected Input ○ InputSignalType ▪ Displayport → DisplayPort ○ OutputFormat ▪ HDMI YUV 444 Limited → YUV 444 Limited ▪ HDMI YUV 422 Limited → YUV 422 Limited HDMI YUV 420 Limited → YUV 420 Limited
1_1_1_0	11/30/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass

```
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='IN1804')
```

SerialOverEthernetClass

```
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='IN1804')
```

SSHClass

#Password Required

```
InterfaceName = ModuleName.SSHClass('192.168.254.254', 22023, Credentials=('admin', 'extron'),  
Model='IN1804')
```

#No Password Required

```
InterfaceName = ModuleName.SSHClass('192.168.254.254', 22023, Credentials=('admin', ''),  
Model='IN1804')
```

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command	Value	Value	
AspectRatio	'Fill'	'Follow'	
Qualifier Key	Qualifier Value		
'Input'	'1' – '4'		
# AspectRatio example InterfaceName.Set('AspectRatio', 'Fill', {'Input': '1'})			
Command	Value	Value	Value
AudioFormat	'Analog'	'LPCM-2Ch'	'Multi-Ch'
	'LPCM-2Ch Auto'	'Multi-Ch Auto'	None
Qualifier Key	Qualifier Value		
'Input'	'1' – '4'		
# AudioFormat example InterfaceName.Set('AudioFormat', 'Analog', {'Input': '1'})			
Command	Value	Value	
AudioMute	'On'	'Off'	
Qualifier Key	Qualifier Value		Qualifier Value
'Output'	'1A'		'1B'
# AudioMute example InterfaceName.Set('AudioMute', 'On', {'Output': '1A'})			
Command	Value	Value	Value
AutoImage	'Execute'	'Execute and Fill'	'Execute and Follow'
# AutoImage example InterfaceName.Set('AutoImage', 'Execute')			
Command	Value	Value	Value
AutoSwitchMode	'User Defined Priority'	'Last Connected Input'	'Off'
# AutoSwitchMode example InterfaceName.Set('AutoSwitchMode', 'User Defined Priority')			
Command	Value	Value	Value
ExecutiveMode ¹	'Off'	'Mode 1'	'Mode 2'
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'Off')			
Command	Value	Value	
Freeze	'On'	'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command	Value	Value	
GlobalAudioMute	'On'	'Off'	
# GlobalAudioMute example InterfaceName.Set('GlobalAudioMute', 'On')			
Command	Value	Value	Value
GlobalVideoMute	'On'	'On with Sync'	'Off'
# GlobalVideoMute example InterfaceName.Set('GlobalVideoMute', 'On')			
Command	Value	Value	
HDCPInputAuthorization	'On'	'Off'	

Qualifier Key 'Input'	Qualifier Value '2' – '4'		
# HDCPInputAuthorization example InterfaceName.Set('HDCPInputAuthorization', 'On', {'Input': '2'})			
Command Input	Value '1' – '4'		
# Input example InterfaceName.Set('Input', '1')			
Command InputGain	Value -18 to 24 in steps of 1		
# InputGain example InterfaceName.Set('InputGain', 24)			
Command InputPresetRecall	Value '1' – '128'		
# InputPresetRecall example InterfaceName.Set('InputPresetRecall', '1')			
Command InputPresetSave	Value '1' – '128'		
# InputPresetSave example InterfaceName.Set('InputPresetSave', '1')			
Command Logo	Value '1' – '16'	Value 'Off'	
# Logo example InterfaceName.Set('Logo', '1')			
Command OutputFormat	Value 'Auto' 'HDMI RGB 444 Limited' 'YUV 420 Limited' ²	Value 'DVI RGB 444' 'YUV 444 Limited'	Value 'HDMI RGB 444 Full' 'YUV 422 Limited'
Qualifier Key 'Output'	Qualifier Value '1A'	Qualifier Value '1B'	
# OutputFormat example InterfaceName.Set('OutputFormat', 'Auto', {'Output': '1A'})			
Command OutputResolution	Value '640x480 (60Hz)' '1280x768 (60Hz)' '1360x768 (60Hz)' '1400x1050 (60Hz)' '1600x1200 (60Hz)' '480p (60Hz)' '720p (29.97Hz)' '720p (59.94Hz)' '1080i (59.94Hz)' '1080p (24Hz)' '1080p (30Hz)' '1080p (60Hz)' '2048x1080 (2K) (25Hz)' '2048x1080 (2K) (50Hz)' '2048x1200 (60Hz)' '2560x1440 (60Hz)'	Value '800x600 (60Hz)' '1280x800 (60Hz)' '1366x768 (60Hz)' '1600x900 (60Hz)' '1920x1200 (60Hz)' '576p (50Hz)' '720p (30Hz)' '720p (60Hz)' '1080i (60Hz)' '1080p (25Hz)' '1080p (50Hz)' '2048x1080 (2K) (23.98Hz)' '2048x1080 (2K) (29.97Hz)' '2048x1080 (2K) (59.94Hz)' '2048x1536 (60Hz)' '2560x1600 (60Hz)'	Value '1024x768 (60Hz)' '1280x1024 (60Hz)' '1440x900 (60Hz)' '1680x1050 (60Hz)' '480p (59.94Hz)' '720p (25Hz)' '720p (50Hz)' '1080i (50Hz)' '1080p (23.98Hz)' '1080p (29.97Hz)' '1080p (59.94Hz)' '2048x1080 (2K) (24Hz)' '2048x1080 (2K) (30Hz)' '2048x1080 (2K) (60Hz)' '2560x1080 (60Hz)' '3840x2160 (23.98Hz)'

	'3840x2160 (24Hz)'	'3840x2160 (25Hz)'	'3840x2160 (29.97Hz)'
	'3840x2160 (30Hz)'	'3840x2160 (50Hz)'	'3840x2160 (59.94Hz)'
	'3840x2160 (60Hz)'	'4096x2160 (23.98Hz)'	'4096x2160 (24Hz)'
	'4096x2160 (25Hz)'	'4096x2160 (29.97Hz)'	'4096x2160 (30Hz)'
	'4096x2160 (50Hz)'	'4096x2160 (59.94Hz)'	'4096x2160 (60Hz)'
	'Custom 1'	'Custom 2'	'Custom 3'
	'Custom 4'	'Custom 5'	
# OutputResolution example InterfaceName.Set('OutputResolution', '640x480 (60Hz)')			
Command	Value	Value	
PowerSaveMode	'On'	'Off'	
# PowerSaveMode example InterfaceName.Set('PowerSaveMode', 'On')			
Command	Value	Value	
Tally	'Open'	'Close'	
Qualifier Key	Qualifier Value		
'Port'	'1' – '7'		
# Tally example InterfaceName.Set('Tally', 'Open', {'Port': '1'})			
Command	Value	Value	Value
TestPattern	'Crop'	'Alternating Pixels'	'Crosshatch'
	'Color Bars'	'Grayscale'	'Audio Test'
	'Off'		
# TestPattern example InterfaceName.Set('TestPattern', 'Crop')			
Command	Value	Value	Value
VideoMute	'On'	'Off'	'On with Sync'
Qualifier Key	Qualifier Value	Qualifier Value	
'Output'	'1A'	'1B'	
# VideoMute example InterfaceName.Set('VideoMute', 'On', {'Output': '1A'})			
Command	Value		
Volume	-100 to 0 in steps of 1		
# Volume example InterfaceName.Set('Volume', 0)			

¹ Mode 1: Lock Front Panel

Mode 2: Limit Front Panel to Input Switching

² Only available when the current HDMI output resolution is 4K/UHD 4096x2160 @ 50, 59.94, or 60 Hz and use half the TMDS character rate.

Status Available

For all commands except for InputSignalType, ScreenSaverStatus, and Temperature, Update should be called only once since the command's status will be updated automatically as the device's status changes.

ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	
AspectRatio	'Fill'	'Follow'	
Qualifier Key	Qualifier Value		
'Input'	'1' – '4'		
# AspectRatio example InterfaceName.Update('AspectRatio', {'Input': '1'}) Value = InterfaceName.ReadStatus('AspectRatio', {'Input': '1'}) InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command	Value	Value	Value
AudioFormat	'Analog'	'LPCM-2Ch'	'Multi-Ch'
	'LPCM-2Ch Auto'	'Multi-Ch Auto'	'None'
Qualifier Key	Qualifier Value		
'Input'	'1' – '4'		
# AudioFormat example InterfaceName.Update('AudioFormat', {'Input': '1'}) Value = InterfaceName.ReadStatus('AudioFormat', {'Input': '1'}) InterfaceName.SubscribeStatus('AudioFormat', None, FeedbackHandler)			
Command	Value	Value	
AudioMute	'On'	'Off'	
Qualifier Key	Qualifier Value	Qualifier Value	
'Output'	'1A'	'1B'	
# AudioMute example InterfaceName.Update('AudioMute', {'Output': '1A'}) Value = InterfaceName.ReadStatus('AudioMute', {'Output': '1A'}) InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command	Value	Value	Value
AutoSwitchMode	'User Defined Priority'	'Last Connected Input'	'Off'
# AutoSwitchMode example InterfaceName.Update('AutoSwitchMode') Value = InterfaceName.ReadStatus('AutoSwitchMode') InterfaceName.SubscribeStatus('AutoSwitchMode', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			

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Command	Value		
ContactStatus	'Open'	'Closed'	
Qualifier Key	Qualifier Value		
'Port'	'1' – '7'		
# ContactStatus example InterfaceName.Update('ContactStatus', {'Port': '1'}) Value = InterfaceName.ReadStatus('ContactStatus', {'Port': '1'}) InterfaceName.SubscribeStatus('ContactStatus', None, FeedbackHandler)			
Command	Value	Value	Value
ExecutiveMode ¹	'Off'	'Mode 1'	'Mode 2'
# ExecutiveMode example InterfaceName.Update('ExecutiveMode') Value = InterfaceName.ReadStatus('ExecutiveMode') InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)			
Command	Value	Value	
Freeze	'On'	'Off'	
# Freeze example InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command	Value	Value	
GlobalAudioMute	'On'	'Off'	
# GlobalAudioMute example InterfaceName.Update('GlobalAudioMute') Value = InterfaceName.ReadStatus('GlobalAudioMute') InterfaceName.SubscribeStatus('GlobalAudioMute', None, FeedbackHandler)			
Command	Value	Value	Value
GlobalVideoMute	'On'	'On with Sync'	'Off'
# GlobalVideoMute example InterfaceName.Update('GlobalVideoMute') Value = InterfaceName.ReadStatus('GlobalVideoMute') InterfaceName.SubscribeStatus('GlobalVideoMute', None, FeedbackHandler)			
Command	Value	Value	
HDCPInputAuthorization	'On'	'Off'	
Qualifier Key	Qualifier Value		
'Input'	'2' – '4'		
# HDCPInputAuthorization example InterfaceName.Update('HDCPInputAuthorization', {'Input': '2'}) Value = InterfaceName.ReadStatus('HDCPInputAuthorization', {'Input': '2'}) InterfaceName.SubscribeStatus('HDCPInputAuthorization', None, FeedbackHandler)			
Command	Value	Value	Value
HDCPInputStatus	'No Source Device Detected'	'Source Detected with HDCP'	'Source Detected without HDCP'
Qualifier Key	Qualifier Value		
'Input'	'1' – '4'		
# HDCPInputStatus example InterfaceName.Update('HDCPInputStatus', {'Input': '1'}) Value = InterfaceName.ReadStatus('HDCPInputStatus', {'Input': '1'}) InterfaceName.SubscribeStatus('HDCPInputStatus', None, FeedbackHandler)			
Command	Value	Value	Value
HDCPOutputStatus	'No Sink Device Detected'	'Sink Detected with HDCP'	'Sink Detected without HDCP'
Qualifier Key	Qualifier Value	Qualifier Value	
'Output'	'1A'	'1B'	
# HDCPOutputStatus example InterfaceName.Update('HDCPOutputStatus', {'Output': '1A'})			

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Value = InterfaceName.ReadStatus('HDCPOutputStatus', {'Output': '1A'}) InterfaceName.SubscribeStatus('HDCPOutputStatus', None, FeedbackHandler)			
Command Input	Value '1' – '4'		
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command InputGain	Value -18 to 24 in steps of 1		
# InputGain example InterfaceName.Update('InputGain') Value = InterfaceName.ReadStatus('InputGain') InterfaceName.SubscribeStatus('InputGain', None, FeedbackHandler)			
Command InputSignalStatus	Value 'Active'	Value 'Not Active'	
Qualifier Key 'Input'	Qualifier Value '1' – '4'		
# InputSignalStatus example InterfaceName.Update('InputSignalStatus', {'Input': '1'}) Value = InterfaceName.ReadStatus('InputSignalStatus', {'Input': '1'}) InterfaceName.SubscribeStatus('InputSignalStatus', None, FeedbackHandler)			
Command InputSignalType	Value 'No Signal' 'DisplayPort'	Value 'DVI'	Value 'HDMI'
Qualifier Key 'Input'	Qualifier Value '1' – '4'		
# InputSignalType example InterfaceName.Update('InputSignalType', {'Input': '1'}) Value = InterfaceName.ReadStatus('InputSignalType', {'Input': '1'}) InterfaceName.SubscribeStatus('InputSignalType', None, FeedbackHandler)			
Command Logo	Value '1' – '16'	Value 'Off'	
# Logo example InterfaceName.Update('Logo') Value = InterfaceName.ReadStatus('Logo') InterfaceName.SubscribeStatus('Logo', None, FeedbackHandler)			
Command OutputFormat	Value 'Auto' 'HDMI RGB 444 Limited' 'YUV 420 Limited' ²	Value 'DVI RGB 444' 'YUV 444 Limited'	Value 'HDMI RGB 444 Full' 'YUV 422 Limited'
Qualifier Key 'Output'	Qualifier Value '1A'	Qualifier Value '1B'	
# OutputFormat example InterfaceName.Update('OutputFormat', {'Output': '1A'}) Value = InterfaceName.ReadStatus('OutputFormat', {'Output': '1A'}) InterfaceName.SubscribeStatus('OutputFormat', None, FeedbackHandler)			
Command OutputResolution	Value '640x480 (60Hz)' '1280x768 (60Hz)' '1360x768 (60Hz)' '1400x1050 (60Hz)' '1600x1200 (60Hz)' '480p (60Hz)' '720p (29.97Hz)'	Value '800x600 (60Hz)' '1280x800 (60Hz)' '1366x768 (60Hz)' '1600x900 (60Hz)' '1920x1200 (60Hz)' '576p (50Hz)' '720p (30Hz)'	Value '1024x768 (60Hz)' '1280x1024 (60Hz)' '1440x900 (60Hz)' '1680x1050 (60Hz)' '480p (59.94Hz)' '720p (25Hz)' '720p (50Hz)'

	'720p (59.94Hz)' '1080i (59.94Hz)' '1080p (24Hz)' '1080p (30Hz)' '1080p (60Hz)' '2048x1080 (2K) (25Hz)' '2048x1080 (2K) (50Hz)' '2048x1200 (60Hz)' '2560x1440 (60Hz)' '3840x2160 (24Hz)' '3840x2160 (30Hz)' '3840x2160 (60Hz)' '4096x2160 (25Hz)' '4096x2160 (50Hz)' 'Custom 1' 'Custom 4'	'720p (60Hz)' '1080i (60Hz)' '1080p (25Hz)' '1080p (50Hz)' '2048x1080 (2K) (23.98Hz)' '2048x1080 (2K) (29.97Hz)' '2048x1080 (2K) (59.94Hz)' '2048x1536 (60Hz)' '2560x1600 (60Hz)' '3840x2160 (25Hz)' '3840x2160 (50Hz)' '4096x2160 (23.98Hz)' '4096x2160 (29.97Hz)' '4096x2160 (59.94Hz)' 'Custom 2' 'Custom 5'	'1080i (50Hz)' '1080p (23.98Hz)' '1080p (29.97Hz)' '1080p (59.94Hz)' '2048x1080 (2K) (24Hz)' '2048x1080 (2K) (30Hz)' '2048x1080 (2K) (60Hz)' '2560x1080 (60Hz)' '3840x2160 (23.98Hz)' '3840x2160 (29.97Hz)' '3840x2160 (59.94Hz)' '4096x2160 (24Hz)' '4096x2160 (30Hz)' '4096x2160 (60Hz)' 'Custom 3'
# OutputResolution example InterfaceName.Update('OutputResolution') Value = InterfaceName.ReadStatus('OutputResolution') InterfaceName.SubscribeStatus('OutputResolution', None, FeedbackHandler)			
Command PowerSaveMode	Value 'On'	Value 'Off'	
# PowerSaveMode example InterfaceName.Update('PowerSaveMode') Value = InterfaceName.ReadStatus('PowerSaveMode') InterfaceName.SubscribeStatus('PowerSaveMode', None, FeedbackHandler)			
Command ScreenSaverStatus	Value 'Active Input Detected; Timer not running'	Value 'No Active Input; Timer running; Output sync enabled'	Value 'No Active Input; Timer expired; Output sync disabled'
# ScreenSaverStatus example InterfaceName.Update('ScreenSaverStatus') Value = InterfaceName.ReadStatus('ScreenSaverStatus') InterfaceName.SubscribeStatus('ScreenSaverStatus', None, FeedbackHandler)			
Command Tally	Value 'Open'	Value 'Close'	
Qualifier Key 'Port'	Qualifier Value '1' – '7'		
# Tally example InterfaceName.Update('Tally', {'Port': '1'}) Value = InterfaceName.ReadStatus('Tally', {'Port': '1'}) InterfaceName.SubscribeStatus('Tally', None, FeedbackHandler)			
Command Temperature	Value —		
# Temperature example InterfaceName.Update('Temperature') Value = InterfaceName.ReadStatus('Temperature') InterfaceName.SubscribeStatus('Temperature', None, FeedbackHandler)			
Command TestPattern	Value 'Crop' 'Color Bars'	Value 'Alternating Pixels' 'Grayscale'	Value 'Crosshatch' 'Audio Test'

	'Off'		
# TestPattern example InterfaceName.Update('TestPattern') Value = InterfaceName.ReadStatus('TestPattern') InterfaceName.SubscribeStatus('TestPattern', None, FeedbackHandler)			
Command VideoMute	Value 'On'	Value 'Off'	Value 'On with Sync'
Qualifier Key 'Output'	Qualifier Value '1A'	Qualifier Value '1B'	
# VideoMute example InterfaceName.Update('VideoMute', {'Output': '1A'}) Value = InterfaceName.ReadStatus('VideoMute', {'Output': '1A'}) InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)			
Command Volume	Value -100 to 0 in steps of 1		
# Volume example InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)			

¹ Mode 1: Lock Front Panel

Mode 2: Limit Front Panel to Input Switching

² Only available when the current HDMI output resolution is 4K/UHD 4096x2160 @ 50, 59.94, or 60 Hz and use half the TMDS character rate.

Cable and Adapter Requirements

Captive Screw to Captive Screw

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

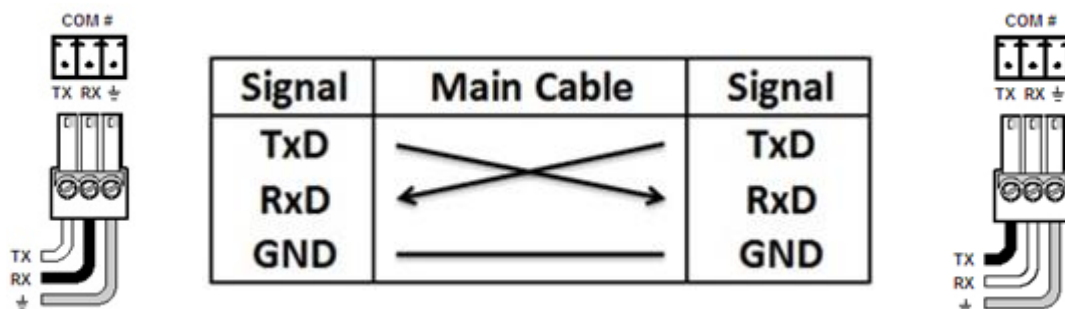
Data Bits: 8

Parity: None

Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet

Default Port: 22023

Logon Credentials Supported: Yes

Default Username: admin

Default Password:

Multi-Connection Yes

Capabilities:

Port Changeability: Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio Fill Input 1	w1*1ASPR\x0D
Aspect Ratio Fill Input 4	w4*1ASPR\x0D
Aspect Ratio Follow Input 1	w1*2ASPR\x0D
Aspect Ratio Follow Input 4	w4*2ASPR\x0D
Audio Format Analog Input 1	wI1*1AFMT\x0D
Audio Format Analog Input 4	wI4*1AFMT\x0D
Audio Format LPCM-2Ch Auto Input 1	wI1*4AFMT\x0D
Audio Format LPCM-2Ch Auto Input 4	wI4*4AFMT\x0D
Audio Format LPCM-2Ch Input 1	wI1*2AFMT\x0D
Audio Format LPCM-2Ch Input 4	wI4*2AFMT\x0D
Audio Format Multi-Ch Auto Input 1	wI1*5AFMT\x0D
Audio Format Multi-Ch Auto Input 4	wI4*5AFMT\x0D
Audio Format Multi-Ch Input 1	wI1*3AFMT\x0D
Audio Format Multi-Ch Input 4	wI4*3AFMT\x0D
Audio Format None Input 1	wI1*0AFMT\x0D
Audio Format None Input 4	wI4*0AFMT\x0D
Audio Mute Off Output 1A	1*0Z
Audio Mute Off Output 1B	2*0Z
Audio Mute On Output 1A	1*1Z
Audio Mute On Output 1B	2*1Z
Auto Image Execute	1*0A
Auto Image Execute and Fill	1*1A
Auto Image Execute and Follow	1*2A
Auto Switch Mode Last Connected Input	w2AUSW\x0D
Auto Switch Mode Off	w0AUSW\x0D
Auto Switch Mode User Defined Priority	w1AUSW\x0D
Executive Mode Mode 1	1X
Executive Mode Mode 2	2X
Executive Mode Off	0X
Freeze Off	1*0F
Freeze On	1*1F
Global Audio Mute Off	0Z
Global Audio Mute On	1Z
Global Video Mute Off	0B
Global Video Mute On	1B
Global Video Mute On with Sync	2B
HDCP Input Authorization Off Input 2	wE2*0HDCP\x0D
HDCP Input Authorization Off Input 4	wE4*0HDCP\x0D
HDCP Input Authorization On Input 2	wE2*1HDCP\x0D
HDCP Input Authorization On Input 4	wE4*1HDCP\x0D
Input 1	1!
Input 4	4!

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Communication Sheet

Input Gain -18	-18G
Input Gain 24	24G
Input Preset Recall 1	2*1.
Input Preset Recall 128	2*128.
Input Preset Save 1	2*1,
Input Preset Save 128	2*128,
Logo 1	wE1*1LOGO\x0D
Logo 16	wE1*16LOGO\x0D
Logo Off	wE1*0LOGO\x0D
Output Format Auto Output 1A	w1*0VTPO\x0D
Output Format Auto Output 1B	w2*0VTPO\x0D
Output Format DVI RGB 444 Output 1A	w1*1VTPO\x0D
Output Format DVI RGB 444 Output 1B	w2*1VTPO\x0D
Output Format HDMI RGB 444 Full Output 1A	w1*2VTPO\x0D
Output Format HDMI RGB 444 Full Output 1B	w2*2VTPO\x0D
Output Format HDMI RGB 444 Limited Output 1A	w1*3VTPO\x0D
Output Format HDMI RGB 444 Limited Output 1B	w2*3VTPO\x0D
Output Format YUV 420 Limited Output 1A	w1*9VTPO\x0D
Output Format YUV 420 Limited Output 1B	w2*9VTPO\x0D
Output Format YUV 422 Limited Output 1A	w1*7VTPO\x0D
Output Format YUV 422 Limited Output 1B	w2*7VTPO\x0D
Output Format YUV 444 Limited Output 1A	w1*5VTPO\x0D
Output Format YUV 444 Limited Output 1B	w2*5VTPO\x0D
Output Resolution 1024x768 (60Hz)	w1*12RATE\x0D\x0A
Output Resolution 1080i (50Hz)	w1*35RATE\x0D\x0A
Output Resolution 1080i (59.94Hz)	w1*36RATE\x0D\x0A
Output Resolution 1080i (60Hz)	w1*37RATE\x0D\x0A
Output Resolution 1080p (23.98Hz)	w1*38RATE\x0D\x0A
Output Resolution 1080p (24Hz)	w1*39RATE\x0D\x0A
Output Resolution 1080p (25Hz)	w1*40RATE\x0D\x0A
Output Resolution 1080p (29.97Hz)	w1*41RATE\x0D\x0A
Output Resolution 1080p (30Hz)	w1*42RATE\x0D\x0A
Output Resolution 1080p (50Hz)	w1*43RATE\x0D\x0A
Output Resolution 1080p (59.94Hz)	w1*44RATE\x0D\x0A
Output Resolution 1080p (60Hz)	w1*45RATE\x0D\x0A
Output Resolution 1280x1024 (60Hz)	w1*15RATE\x0D\x0A
Output Resolution 1280x768 (60Hz)	w1*13RATE\x0D\x0A
Output Resolution 1280x800 (60Hz)	w1*14RATE\x0D\x0A
Output Resolution 1360x768 (60Hz)	w1*16RATE\x0D\x0A
Output Resolution 1366x768 (60Hz)	w1*17RATE\x0D\x0A
Output Resolution 1400x1050 (60Hz)	w1*19RATE\x0D\x0A
Output Resolution 1440x900 (60Hz)	w1*18RATE\x0D\x0A
Output Resolution 1600x1200 (60Hz)	w1*22RATE\x0D\x0A
Output Resolution 1600x900 (60Hz)	w1*20RATE\x0D\x0A

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Output Resolution 1680x1050 (60Hz)	w1*21RATE\x0D\x0A
Output Resolution 1920x1200 (60Hz)	w1*23RATE\x0D\x0A
Output Resolution 2048x1080 (2K) (23.98Hz)	w1*46RATE\x0D\x0A
Output Resolution 2048x1080 (2K) (24Hz)	w1*47RATE\x0D\x0A
Output Resolution 2048x1080 (2K) (25Hz)	w1*48RATE\x0D\x0A
Output Resolution 2048x1080 (2K) (29.97Hz)	w1*49RATE\x0D\x0A
Output Resolution 2048x1080 (2K) (30Hz)	w1*50RATE\x0D\x0A
Output Resolution 2048x1080 (2K) (50Hz)	w1*51RATE\x0D\x0A
Output Resolution 2048x1080 (2K) (59.94Hz)	w1*52RATE\x0D\x0A
Output Resolution 2048x1080 (2K) (60Hz)	w1*53RATE\x0D\x0A
Output Resolution 2048x1200 (60Hz)	w1*54RATE\x0D\x0A
Output Resolution 2048x1536 (60Hz)	w1*55RATE\x0D\x0A
Output Resolution 2560x1080 (60Hz)	w1*56RATE\x0D\x0A
Output Resolution 2560x1440 (60Hz)	w1*57RATE\x0D\x0A
Output Resolution 2560x1600 (60Hz)	w1*58RATE\x0D\x0A
Output Resolution 3840x2160 (23.98Hz)	w1*59RATE\x0D\x0A
Output Resolution 3840x2160 (24Hz)	w1*60RATE\x0D\x0A
Output Resolution 3840x2160 (25Hz)	w1*61RATE\x0D\x0A
Output Resolution 3840x2160 (29.97Hz)	w1*62RATE\x0D\x0A
Output Resolution 3840x2160 (30Hz)	w1*63RATE\x0D\x0A
Output Resolution 3840x2160 (50Hz)	w1*64RATE\x0D\x0A
Output Resolution 3840x2160 (59.94Hz)	w1*65RATE\x0D\x0A
Output Resolution 3840x2160 (60Hz)	w1*66RATE\x0D\x0A
Output Resolution 4096x2160 (23.98Hz)	w1*69RATE\x0D\x0A
Output Resolution 4096x2160 (24Hz)	w1*70RATE\x0D\x0A
Output Resolution 4096x2160 (25Hz)	w1*71RATE\x0D\x0A
Output Resolution 4096x2160 (29.97Hz)	w1*72RATE\x0D\x0A
Output Resolution 4096x2160 (30Hz)	w1*73RATE\x0D\x0A
Output Resolution 4096x2160 (50Hz)	w1*74RATE\x0D\x0A
Output Resolution 4096x2160 (59.94Hz)	w1*75RATE\x0D\x0A
Output Resolution 4096x2160 (60Hz)	w1*76RATE\x0D\x0A
Output Resolution 480p (59.94Hz)	w1*24RATE\x0D\x0A
Output Resolution 480p (60Hz)	w1*25RATE\x0D\x0A
Output Resolution 576p (50Hz)	w1*26RATE\x0D\x0A
Output Resolution 640x480 (60Hz)	w1*10RATE\x0D\x0A
Output Resolution 720p (25Hz)	w1*29RATE\x0D\x0A
Output Resolution 720p (29.97Hz)	w1*30RATE\x0D\x0A
Output Resolution 720p (30Hz)	w1*31RATE\x0D\x0A
Output Resolution 720p (50Hz)	w1*32RATE\x0D\x0A
Output Resolution 720p (59.94Hz)	w1*33RATE\x0D\x0A
Output Resolution 720p (60Hz)	w1*34RATE\x0D\x0A
Output Resolution 800x600 (60Hz)	w1*11RATE\x0D\x0A
Output Resolution Custom 1	w1*201RATE\x0D\x0A
Output Resolution Custom 2	w1*202RATE\x0D\x0A

**Global Scripter Module
Communication Sheet**

Output Resolution Custom 3	w1*203RATE\x0D\x0A
Output Resolution Custom 4	w1*204RATE\x0D\x0A
Output Resolution Custom 5	w1*205RATE\x0D\x0A
Power Save Mode Off	w0PSAV\x0D
Power Save Mode On	w1PSAV\x0D
Tally Close Port 1	w1*1TALY\x0D
Tally Close Port 7	w7*1TALY\x0D
Tally Open Port 1	w1*0TALY\x0D
Tally Open Port 7	w7*0TALY\x0D
Test Pattern Alternating Pixels	w1*2TEST\x0D
Test Pattern Audio Test	w1*6TEST\x0D
Test Pattern Color Bars	w1*4TEST\x0D
Test Pattern Crop	w1*1TEST\x0D
Test Pattern Crosshatch	w1*3TEST\x0D
Test Pattern Grayscale	w1*5TEST\x0D
Test Pattern Off	w1*0TEST\x0D
Video Mute Off Output 1A	1*0B
Video Mute Off Output 1B	2*0B
Video Mute On Output 1A	1*1B
Video Mute On Output 1B	2*1B
Video Mute On with Sync Output 1A	1*2B
Video Mute On with Sync Output 1B	2*2B
Volume 0	0V
Volume -100	-100V

Appendix B. Update Commands

Aspect Ratio Input 1	w1ASPR\x0D
Aspect Ratio Input 4	w4ASPR\x0D
Audio Format Input 1	wI1AFMT\x0D
Audio Format Input 4	wI4AFMT\x0D
Audio Mute Output 1A	1*Z
Audio Mute Output 1B	2*Z
Auto Switch Mode	wAUSW\x0D
Contact Status Port 1	wCNTC\x0D
Contact Status Port 7	wCNTC\x0D
Executive Mode	X
Freeze	1F
Global Audio Mute	Z
Global Video Mute	B
HDCP Input Authorization Input 2	wE2HDCP\x0D
HDCP Input Authorization Input 4	wE4HDCP\x0D
HDCP Input Status Input 1	wI1HDCP\x0D
HDCP Input Status Input 4	wI4HDCP\x0D
HDCP Output Status Output 1A	wO1HDCP\x0D
HDCP Output Status Output 1B	wO2HDCP\x0D
Input	!
Input Gain	G
Input Signal Status Input 1	w0LS\x0D
Input Signal Status Input 4	w0LS\x0D
Input Signal Type Input 1	1*\
Input Signal Type Input 4	4*\
Logo	wE1LOGO\x0D
Output Format Output 1A	w1*VTPO\x0D
Output Format Output 1B	w2*VTPO\x0D
Output Resolution	w1RATE\x0D
Power Save Mode	wPSAV\x0D
Screen Saver Status	wS1SSAV\x0D
Tally Port 1	wTALY\x0D
Tally Port 7	wTALY\x0D
Temperature	w20STAT\x0D
Test Pattern	w1TEST\x0D
Video Mute Output 1A	B
Video Mute Output 1B	B
Volume	V